

Scale Fixes Local Errors First: Differential Scaling of Chess Transformer Failures

Authors and affiliations

Mohit Pratap Singh Rathore

Ovigo Private Limited, India

Gunveer Sinh Kalsi

Ovigo Private Limited, India

Sriharsha Meduri

Ovigo Private Limited, India

Corresponding author. Sriharsha Meduri, Ovigo Private Limited, India. Email: sriharshameduri07@gmail.com

Word count. Approximately 6,280 words in the body, excluding the abstract, references, and declarations.

Declarations

Disclosure statement. The authors report no conflict of interest.

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability statement. The analysis code, every result file cited by the manuscript, and the macro provenance record are available at <https://github.com/MohitPratapSinghRathore/derived-total>. Trained checkpoints are available from the corresponding author on reasonable request, being too large to distribute with the repository. External checkpoints are public.

AI usage declaration. During the preparation of this manuscript the authors used a generative AI tool to assist with language editing and refinement of the text, and with writing analysis code. The authors reviewed and edited all output and take full responsibility for the content of the paper. Generative AI was not used to generate or analyse the study’s data, which were produced entirely by the measurement pipeline described in the paper.

Contributor biographies

Mohit Pratap Singh Rathore is a researcher at Ovigo Private Limited, where his work concerns the measurement of internal state in sequence models and the evaluation of long-horizon failure.

Gunveer Sinh Kalsi is a researcher at Ovigo Private Limited, working on model evaluation infrastructure and the reproducibility of measured claims.

Sriharsha Meduri is a researcher at Oviqo Private Limited. His research interests include the design and evaluation of decision-support artifacts and the governance of machine learning systems in organisational settings.